

Brandon Li

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EDUCATION

University of California, Los Angeles (UCLA)

Bachelor of Science in Computer Science

Expected June 2028

Los Angeles, CA

EXPERIENCE

Software Development Engineer Intern

Sep 2026 – Present

Amazon (Amazon Web Services)

Boston, MA

- Extended Amazon's Remote MCP server with 8 endpoints by integrating an internal work-management service (similar to Jira), enabling hundreds of thousands of employees to execute workflows directly from a chatbot.
- Built specialized AI agents for internal work-management workflows, allowing agent-to-agent (A2A) coordination for custom tasks such as summarization and board updates while handling thousands of production requests.
- Implemented A2A authentication and support for user-mocking agents for secure agent interactions and handoffs.
- Improved production reliability and debuggability by adding logging and over a hundred unit, wiring, and deployment tests, catching possible failures before release and validating end-to-end deployment behavior.

Software Engineering Intern

June 2026 – Sep 2026

Google (YouTube)

San Bruno, CA

- Engineered 10-bit software AV1 HDR video playback for the YouTube Android app via low-level ARM/NEON SIMD vectorization in C++, improving the video quality and playback performance for millions of users.
- Defined per-device video resolution caps (e.g. 1080p) for dav1d software decoding, filtering higher-resolution formats out of the playback selection to prevent thermal throttling and battery drain on lower-end devices.
- Built a Java-based shader effect debug UI and OpenGL video pipeline to accelerate developer iteration speed.
- Integrated custom fragment shaders to allow video effects across thousands of DRM-protected shows and movies.
- Resolved memory leaks, migrated deprecated code, and improved format filtering across ExoPlayer and YouTube.

Software Engineering Intern

June 2025 – Sep 2025

VideoLAN

Remote

- Built a shared audio dynamics core, cutting duplicate code by 20% and binary size by 5% to speed up downloads.
- Added ggml/OpenCV to VLC, ported 3 modules to C++, and upgraded to SAM3 to support text segmentation.
- Designed SAM2 object segmentation and YuNet face detection to be able to track over 100 concurrent targets.
- Implemented 5 different dithering algorithms to allow users to convert videos into GIF files at varying qualities.

PROJECTS

yt-media-storage | C++, Assembly/SIMD, FFmpeg, Coding Theory, Compression

Feb 2026 – Present

- Built a C++ tool (910 stars, 110+ forks) that can convert desktop files into YouTube videos for storage.
- Implemented SIMD intrinsics, OpenMP parallelization, and cache optimizations to speed up conversion by 100x.
- Designed a lightweight UI using Qt 6 and added libsodium encryption to enhance usability and security.
- Used CRC and Wirehair error correction to recover data corrupted by lossy video compression algorithms.
- Reached #5 on Hacker News (Y Combinator) in 12 hours, drawing thousands of views and new contributors.
- Published a [technical explanation video](#) on YouTube with over 3.4M impressions, educating over 280k viewers.

mcav | Java, Spring Boot, TypeScript, CI/CD

June 2020 – Present

- Built a 12-module, 150K-line Java media framework (130+ stars, 10 forks) whose unified player API streams real-time video from VLC, OpenCV, Chromium, VNC, and QEMU sources across four pluggable render backends.
- Designed a pure-GPU (GLSL fragment shader) video codec with O(1) per-pixel random access, improving BD-rate over 40% with VMAF-scored optimization rounds to reach 10 ms/frame at 1080p for integrated graphics cards.
- Engineered a low-latency pipeline API with nearly 50 different documented video effects that can be applied.
- Self-hosted a TeamCity CI/CD pipeline that publishes releases to a Maven repository, gated with 2,500+ tests.

TECHNICAL SKILLS

Languages: Java, C/C++, OCaml, JavaScript/TypeScript, Assembly

Frameworks & Libraries: Spring, Hibernate, FFmpeg, OpenCV, libVLC, ExoPlayer, React, Next.js, Tailwind

Technologies: AWS, AI Agents, MCP, Git, CMake, Meson, OpenGL, Android, GLSL, VMAF, AV1, NEON/SIMD

Awards: Dean's Honor List